AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 24 · STUDENT LESSON

Feedback Loops

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

Log what was shown. Log what happened. Inspect the loop.

Explain how a policy shapes future evidence without equating unshown items, nonclicks and dislike.

Keep one habit: Scripted event contract → repeated selection → planned exposure comparison → fresh independent audit.

MAIN LESSON · SECTION 1

Did the item get no clicks, or no chance to be seen?

> Trace how selection changes exposure and the evidence used next.

The current feedback-loop goals

I can separate exposure, clicks and explicit preference feedback.

I can trace how a selection policy changes later evidence.

I can limit conclusions from unshown items and small exposure counts.

Recall a hard eligibility requirement

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

- An option violates a hard duration limit but has the best score. Should it win?

Your end goal

By the end, I can explain how recommendation **exposure** can create a **feedback loop**.

▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Recommendation Scores. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Similarity by Selected Traits. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Missing Data. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your **explanation** that the **evidence** shows was wrong.

Week 24: **Feedback Loop**s

Recommendations influence behavior, which then becomes new data.



THINKER MISSION

Map how repeated clicks can narrow future suggestions.

Thinker name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

This week’s end goal

MAIN LESSON · SECTION 2

Define exposure and feedback before updating the policy

> The simulation makes its assumptions explicit.

The loop and the unshown-item problem

A feedback loop occurs when today’s recommendation changes what can be observed, and those observations influence tomorrow’s recommendation. Suppose a system displays only the item with the most clicks, breaking an initial tie by ID. The first choice can receive every display opportunity and collect more clicks, while an unshown item stays at zero. That zero is not a recorded dislike.

The complete two-book event contract

Use two fictional books A and B. Start with zero displays and zero clicks for each. One displayed slot counts as one exposure; a click is a separate event. In this explicitly invented simulation, an A display receives a click and a B display does not. These scripted events are teaching assumptions, not observations of real readers. Compare a click-count selection policy with a predeclared A,B,A exposure plan; keep the event assumption fixed.

▼ **Earlier wording, alternatives and picture transcripts**

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



MYSTERY

The learners must use **feedback loop**s to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

Build a fictional activity **recommender** using last week's score. Require duration at most a supplied limit. On a tie choose the alphabetically first name.

This makes the outcome reproducible rather than dependent on **input** order. Assume unique names and validated numeric fields.

MAIN LESSON · SECTION 3

Trace selection, display, click and the next selection

> Keep the observation mechanism beside the resulting counts.

Predict and run the complete simulation

```
def simulate(plan=None):
    exposures = {"A": 0, "B": 0}
    clicks = {"A": 0, "B": 0}
    scripted_click = {"A": True, "B": False}
    log = []
    for round_number in range(3):
        if plan is None:
            shown = sorted(clicks, key=lambda name: (-clicks[name], name))[0]
        else:
            shown = plan[round_number]
        exposures[shown] += 1
        clicked = scripted_click[shown]
        if clicked:
```

```
            clicks[shown] += 1
        log.append((round_number + 1, shown, clicked, exposures.copy(),
clicks.copy()))
    return log, exposures, clicks
for label, plan in [("click-count policy", None), ("planned exposure", ["A", "B",
"A"])]:
    log, exposures, clicks = simulate(plan)
    print(label)
    for event in log:
        print(event)
    print("displays", exposures, "clicks", clicks)
```

Explain selection, saved records and scripted evidence

simulate keeps display and click dictionaries separate. The sorting key uses negative clicks to place larger counts first, then ID for equal counts. A supplied plan overrides that selection order without changing the scripted per-book events. copy saves each round’s counts instead of letting later updates overwrite earlier log entries. The returned log records round, shown ID, click event, displays and clicks. No hidden reader preference is inferred or fitted.

▼ Review · Earlier deterministic scoring-function implementation

```
def recommend(items, limit):
    best = None
    best_score = None
    for item in items:
        if item["minutes"] <= limit:
            score = 2 * item["fun"] + item["learning"]
            if (best is None or score > best_score
                or (score == best_score and item["name"] < best)):
                best = item["name"]
                best_score = score
    return best

items = [{"name": "B", "fun": 3, "learning": 2, "minutes": 10},
         {"name": "A", "fun": 2, "learning": 4, "minutes": 9}]
print(recommend(items, 10))
```

Follow the evidence

STEP OR CASE	EVIDENCE	MEANING
B	Eligible score 8	Temporary best B
A	Eligible score 8	Alphabetical tie gives A

Read each row from left to right and connect the evidence to the stated rule.

The function prints nothing; the caller prints A. best starts as None to mean no eligible option found. Python short-circuits or: when best is None, later comparisons in that condition need not run.

After selecting an item, best_score is numeric. An empty list or no eligible item returns None.

Write a trace table with item, eligible, score and best-so-far. In the worked list, B qualifies and scores 8, so B becomes the current best. A also qualifies and scores 8.

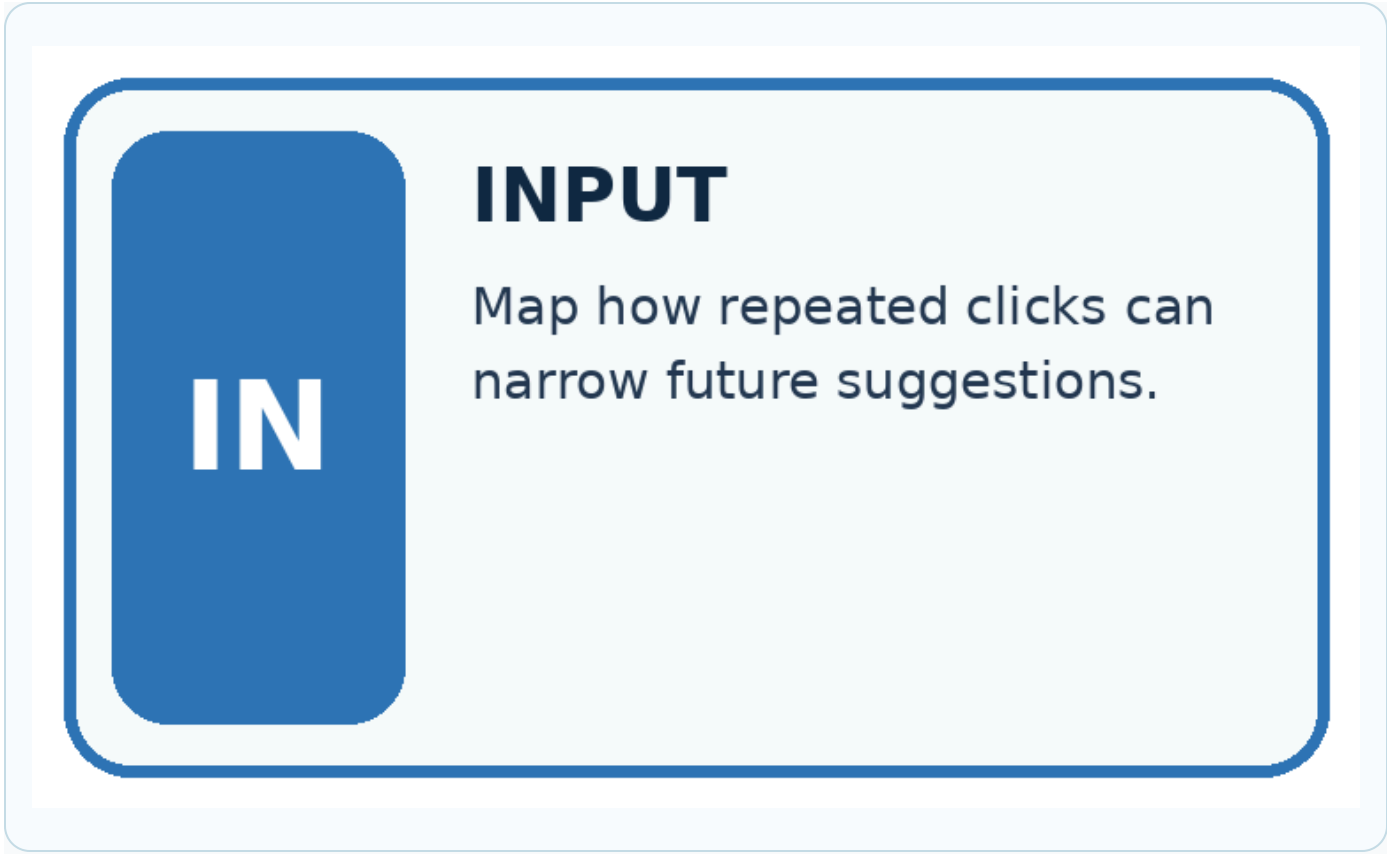
The scores tie, so the written alphabetical rule changes the best to A. Starting with A gives the same final answer. If no item qualifies, best stays None; do not invent a winner or quietly relax the limit.

▼ Earlier wording, alternatives and picture transcripts

VISUAL **MECHANISM**

Input → **Process** → **Output**

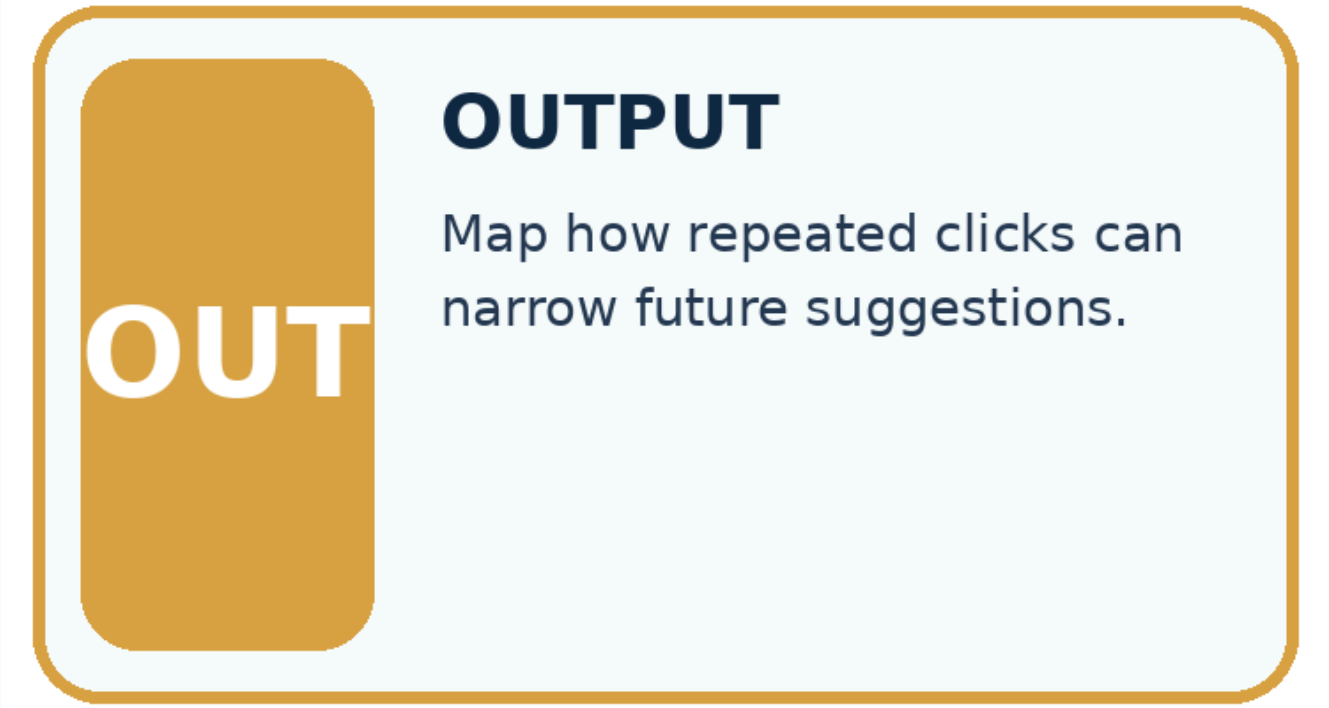
Trace the information. At each stage, ask what changed and how you could test it.


IN
INPUT
Map how repeated clicks can narrow future suggestions.

↓


↻
PROCESS
Recommendations influence behavior, which then becomes new data.

↓



Which stage could fail even when the other two are correct? Explain:

> **Write or draw your reasoning and evidence.**

A closer explanation

Trace the best-so-far record

Connect the picture and the explanation

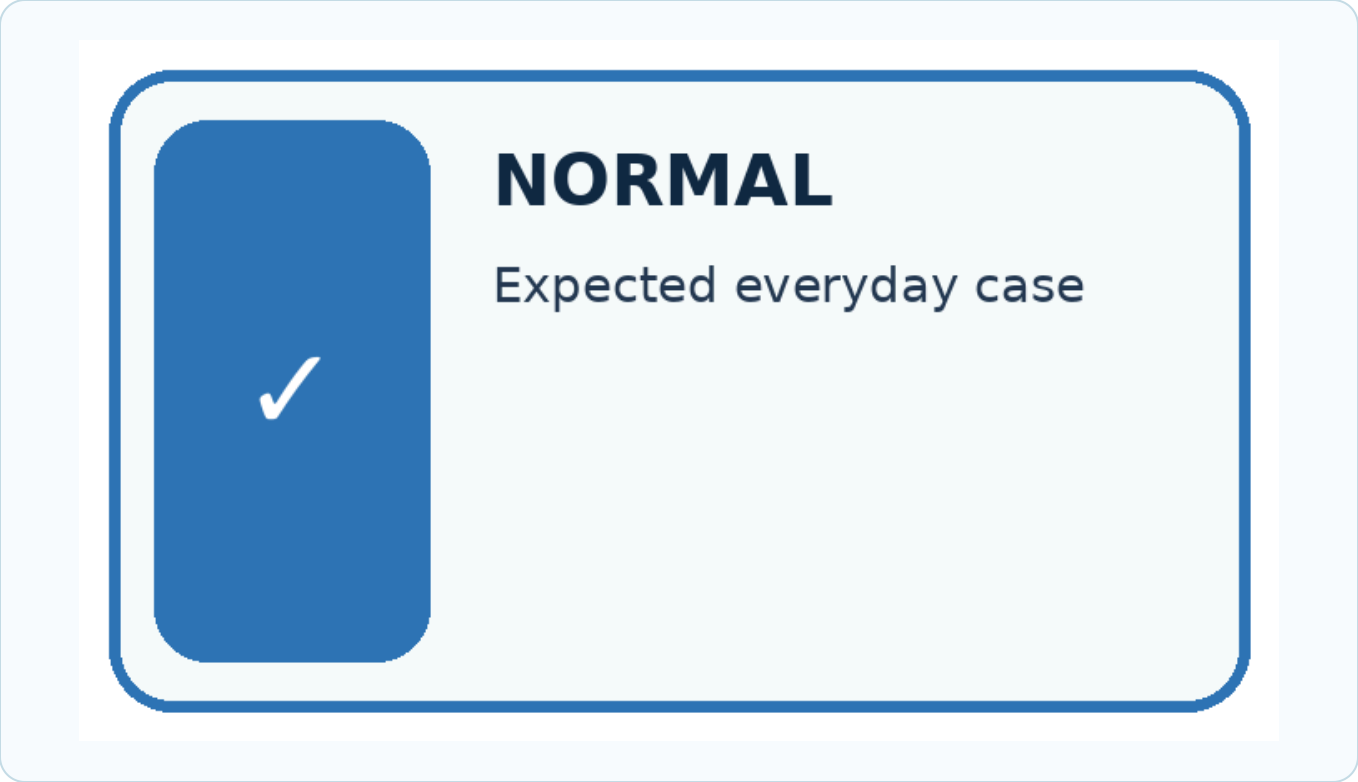
Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

MAIN LESSON · SECTION 4

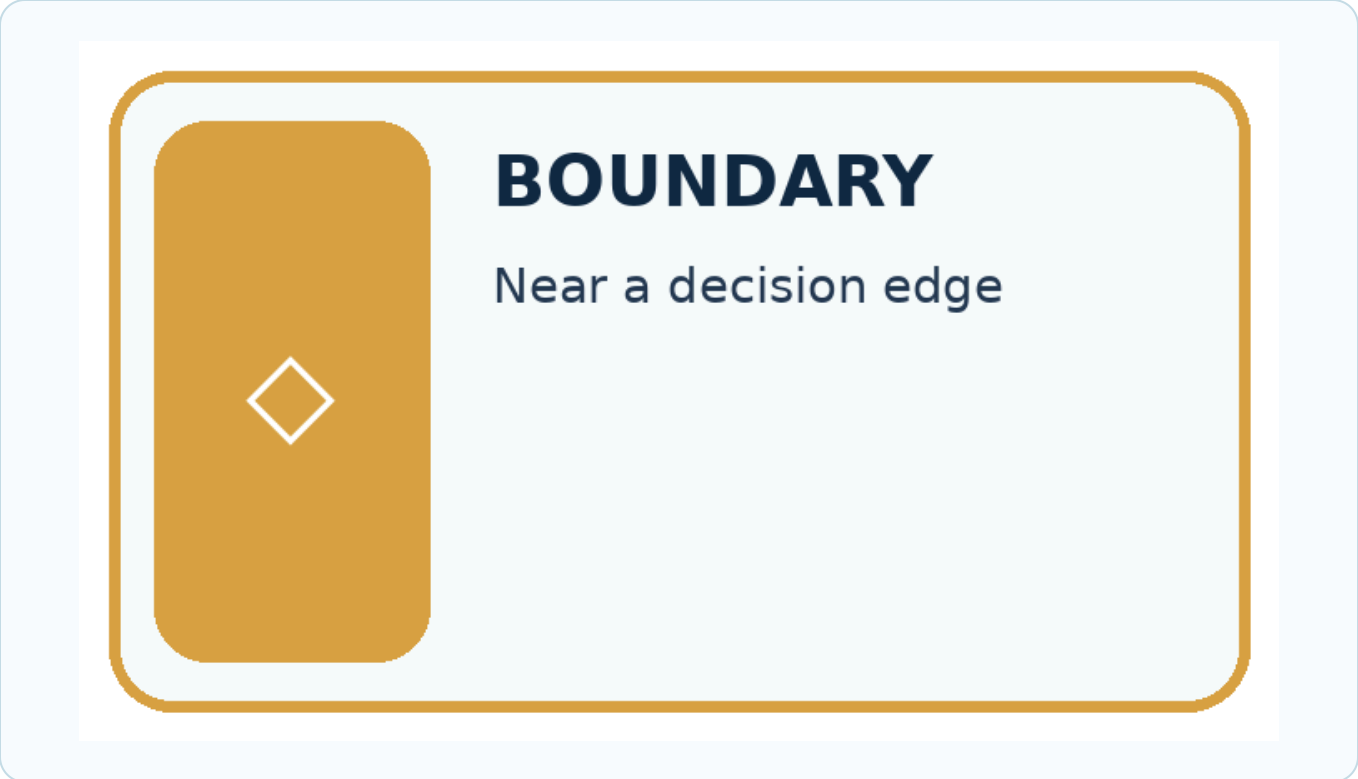
Distinguish not shown from shown without a click

> Neither event alone establishes a permanent preference.

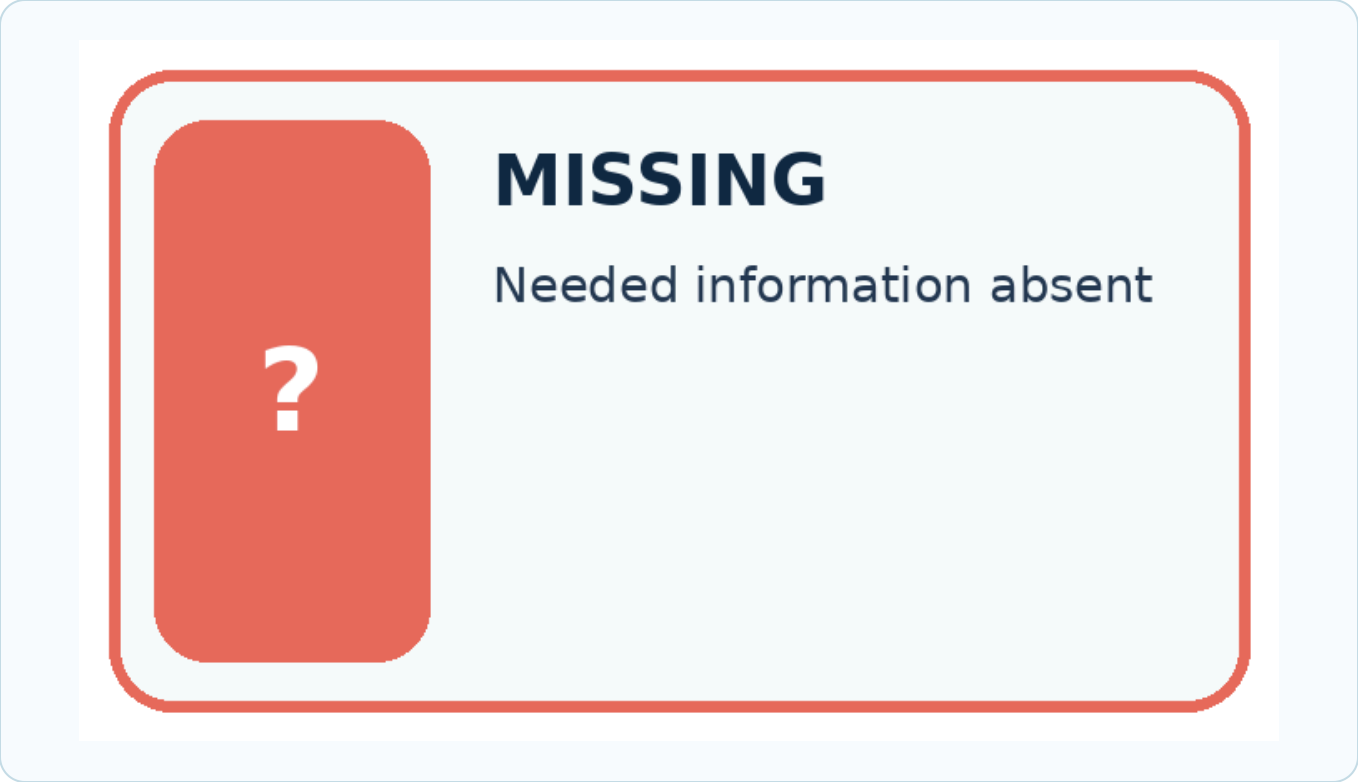
Four pictures with a purpose



Normal case: predict the result and deciding evidence.



Boundary case: identify the exact limit.



Missing information: name what must be clarified.



Transfer case: explain which procedure still applies.

Choose a revealing case

- Which case is most likely to reveal a hidden assumption? Why?

> **Write or draw your reasoning and evidence.**

Predict the supplied changed case

- Change: On a planned trial, show B once and record no click.

<details>
<summary>Check after predicting</summary>

Prediction: B gains one **exposure** but still zero clicks.

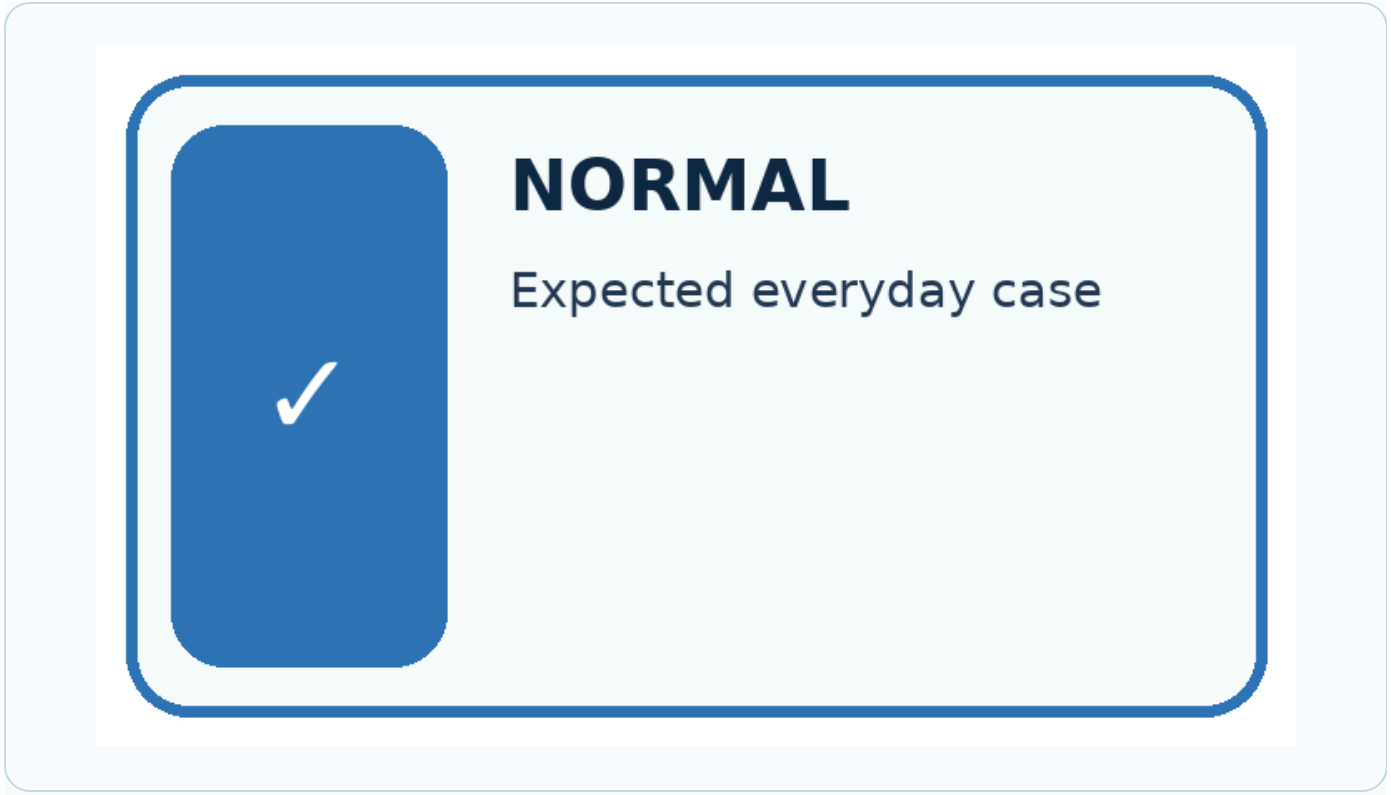
Reason: The **observation** now supports a nonclick on that display, not a permanent dislike.
</details>

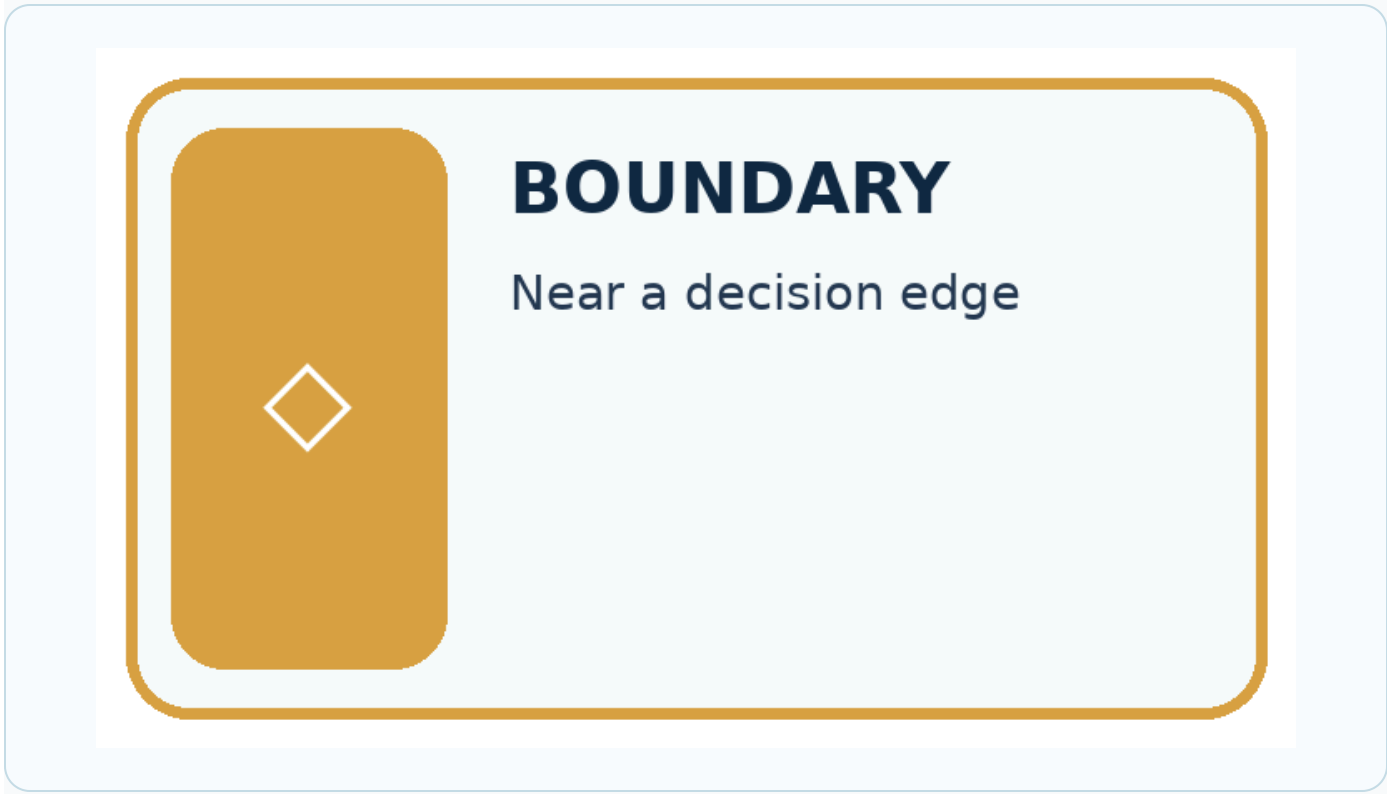
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<summary>Earlier wording, alternatives and picture transcripts</summary>

CASE GALLERY

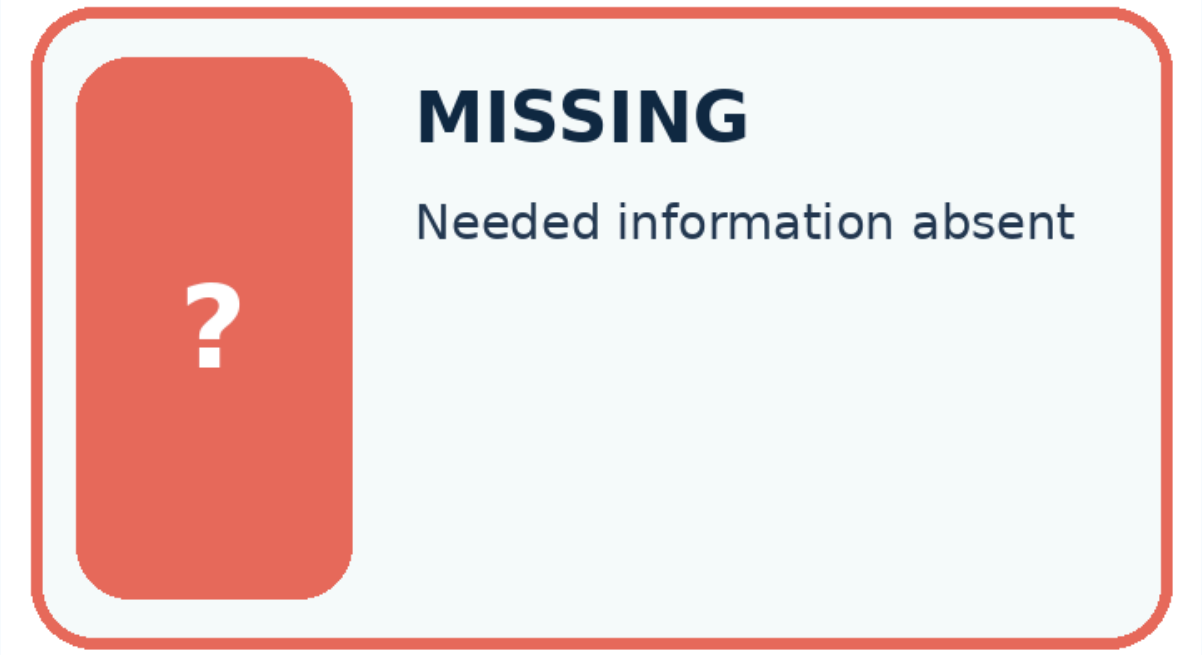
Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.
</details>





NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Follow the three-round click-count loop

> A’s higher count partly reflects the opportunities the policy created.

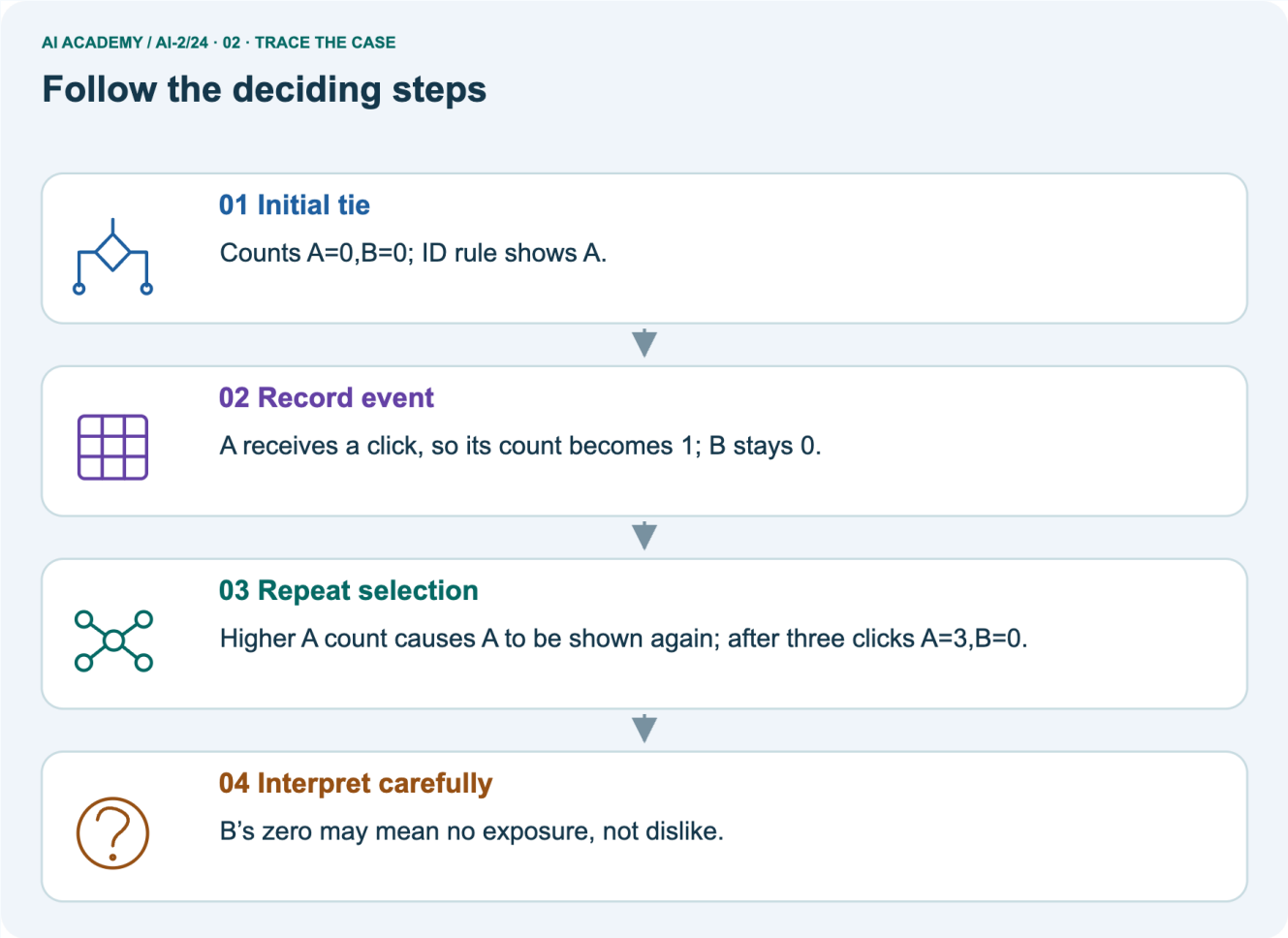
Read the initial tie and invented three-round events

A fictional catalogue repeatedly recommends two books using observed clicks.

Initially each click count is 0. Equal counts choose earlier ID A over B.

Only the recommended book is shown. In an invented three-round record, A is shown and clicked each round; B is never shown.

Trace repeated selection and missing exposure



Follow the labelled picture with the matching explanation. Open the vector diagram

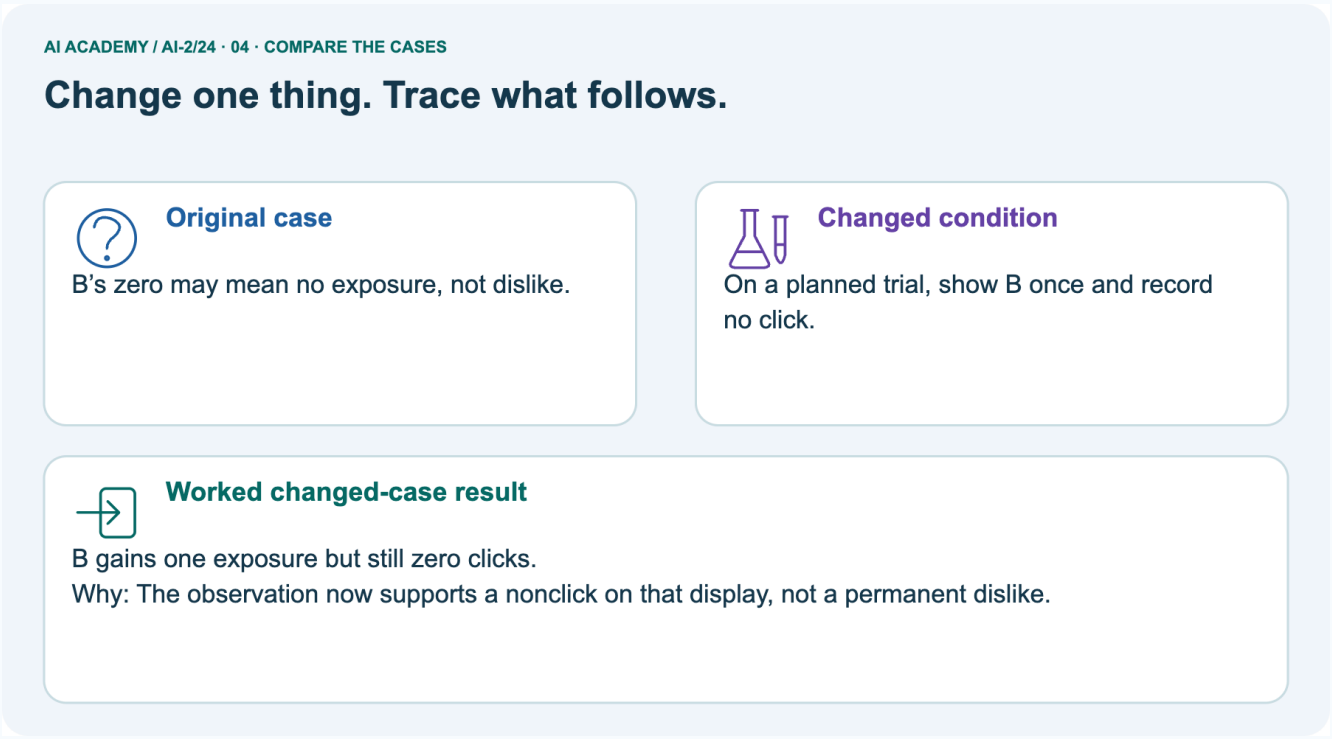
The selection policy changes which **evidence** is available. A accumulates clicks partly because the system gives it every opportunity, while B has none. These records cannot compare enjoyment of two equally exposed books.

Keep display counts separate from click counts; investigate alternatives with a teacher-approved **exposure** plan rather than treating zero as negative feedback.

Predict a planned B display with no click

- Change one thing: On a planned trial, show B once and record no click. Predict the result before checking the **explanation**. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain the needed field and evidence limits

- Read the labelled picture: What extra field would reveal why comparing raw clicks is misleading?
- A click can reflect curiosity or a mistake; even equal **exposure** does not reveal every reason for selection.

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

WEEK 24 • WORKED EVIDENCE

1. System recommends mostly fantasy.
2. Reader can only click among shown books.
3. Clicks are treated as stronger fantasy preference.
4. Next round shows even more fantasy, hiding other possible interests.

1. System recommends mostly fantasy.

2. Reader can only click among shown books.

3. Clicks are treated as stronger fantasy preference.

4. Next round shows even more fantasy, hiding other possible interests.

Recalculate, retrace, or restate the reasoning in your own representation:

> **Write or draw your reasoning and evidence.**

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

> **Predict the result if the list order is reversed, and if the limit is 8.**

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Feedback Loops

The facts we will use

AI ACADEMY / AI-2/24

Exposure and clicks after the invented three rounds

	Shown	Clicked
A	3	3
B	0	0

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/24

Feedback Loops

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Initial tie

Counts A=0,B=0; ID rule shows A.



2. Record event

A receives a click, so its count becomes 1; B stays 0.



3. Repeat selection

Higher A count causes A to be shown again; after three clicks A=3,B=0.



4. Interpret carefully

B's zero may mean no exposure, not dislike.

CHANGE ONE THING

On a planned trial, show B once and record no click.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Initial tie	Counts A=0,B=0; ID rule shows A.
2. Record event	A receives a click, so its count becomes 1; B stays 0.
3. Repeat selection	Higher A count causes A to be shown again; after three clicks A=3,B=0.
4. Interpret carefully	B's zero may mean no exposure, not dislike.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Raw clicks cannot reveal preferences independently of exposure

> Separate observed rates, missing opportunities and interpretation.

Read the denominator and meaning limits

Report exposure counts with click counts. A click can reflect curiosity, position or a mistake, and a nonclick is not an explicit statement of dislike. A zero-click item with zero displays has no observed click rate;0/0 is unavailable. One nonclick after one display gives0/1 for that observation, not a reliable future preference estimate. A controlled exposure plan can broaden opportunities without proving satisfaction or a causal effect by itself.

▼ Review · Earlier score explanation

The recommendation follows a written scoring policy. It does not learn preferences, prove learning benefit or authorize an action. Display the chosen rule and let a person review whether an option is suitable.

> **Write a one-sentence explanation of X winning without calling it objectively best.**

▼ Earlier wording, alternatives and picture transcripts

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: Clicks reveal complete preference, even for items never shown.
Strong move: locate evidence, change one cause, and retest.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Compare logs while keeping the event assumption fixed

> A planned opportunity and its observed outcome are separate fields.

Practise the two complete simulation records

From the original three-round log, list displays and clicks for A and B separately. Then use the supplied A,B,A plan under the same scripted responses. Compare distinct displayed items and each denominator. The plan changes who gets seen; the event rule is not a measurement of what readers actually enjoy. Explain why copying the winning item into every slot would preserve the exposure problem.

Explain the pictured exposure field

What extra field would reveal why comparing raw clicks is misleading?

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Review · Earlier order-dependent scoring repair

Updating best on score >= best_score makes the last tied item win, so reordering can change results. Use the explicit name comparison for ties. Test both orders and a no-match case after the repair.

Use workbook W2 for the connected example “Implementing a **recommender**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own **explanation**.

Use the worked case for Implementing a **recommender**. Proposed **explanation**: “A successful run proves the conclusion”. Decide whether this **explanation** is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

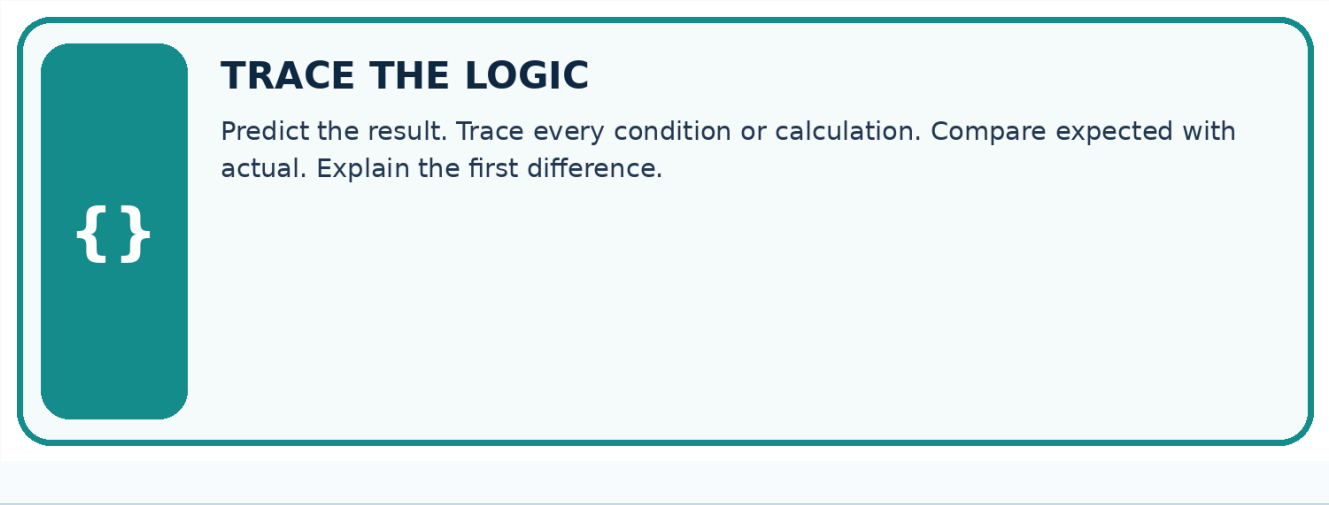
Answer on your own before discussion. If you are unsure, say which step you need help with.

▼ **Earlier wording, alternatives and picture transcripts**

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



SET **evidence**_record PREDICT expected_result RUN one normal case and one boundary case COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST EXPLAIN result, limitation, and next safe step

Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Practise with less help: Handle no match

Check your reasoning

Explain the picture

> **My explanation using a pictured step and the given facts**

MAIN LESSON · SECTION 9

Audit both supplied three-round exposure policies

> Keep predictions, simulated events and unobserved preferences distinct.

Predict, execute, log and compare

1. Before running, predict three shown IDs and final display/click counts for each supplied policy. Use the complete Section3 program or label a paper simulation. Keep every round's choice and count snapshot. Compare diversity as the number of distinct displayed IDs: this small comparison does not measure learning benefit or satisfaction. Record which change affected selection and which responses were scripted. Propose an alternative predeclared exposure plan and explain what new evidence it might allow without inventing real reader reactions. Keep the new X/Y/Z count audit for independent W4.

Keep evidence and explain the observation limit

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

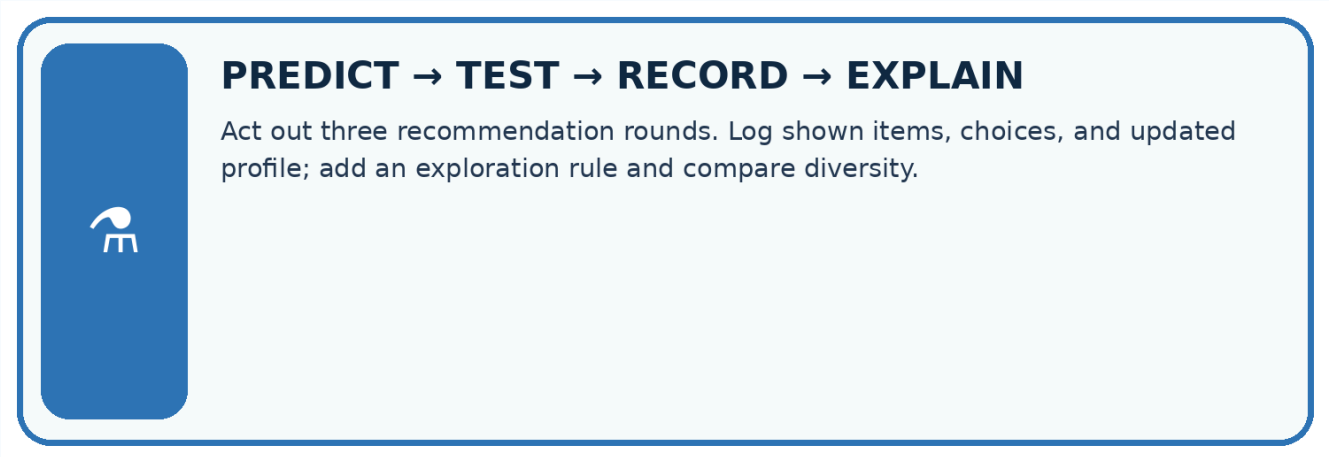
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Use the supplied three-round click-count policy and A,B,A exposure plan under the same scripted events. Log displays and clicks separately and compare distinct displayed IDs. Do not claim real preference or satisfaction was measured.

Plan → try → check
</details>

MAIN LESSON · SECTION 14

Try alone: audit the fresh X/Y/Z exposure log

> Keep first-attempt reasoning before hints or the teacher key.

Current independent task and new response

- A fresh fictional log has X displayed4 times with1 click, Y displayed4 times with2 clicks, and Z displayed0 times with0 clicks. Calculate the observed click rates for X and Y; state why Z has no measured click rate. Explain how always selecting the highest raw-click item can shape future evidence. Then a planned trial shows Z once with no click: update Z's counts and rate, and state one limit of that observation.

My independent exposure-log reasoning

Fair attempt and feedback

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Review · Earlier implementation task and saved response**

Independent case: Implement independently

Use the function with X (fun 4, learning 1, minutes 12), Y (3,3,10), Z (5,5,13), limit 12. Predict the winner, reverse the list and retest. Also test limit 9.

My independent explanation, trace and limits

▼ **Earlier wording, alternatives and picture transcripts**

Week 24 Independent application

Close the worked **explanation**. Apply the idea to this different question without copying.

Only space books are shown, so nearly every click is on space. Can these clicks prove readers dislike other topics? Explain the loop.

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Reserve some varied suggestions and distinguish ‘not shown’ from ‘not liked.’

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

A **recommender** shows poetry in 2 slots and adventure in 20. Poetry gets 1 click; adventure gets 6. Compute clicks per shown slot and explain why neither rate alone proves preference.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Record the evidence your recommender creates as well as uses

> Inspect whether the current policy narrows later feedback.

This week’s concrete contribution

For the fictional recommender, retain the displayed options, slot or occasion, selected option, clicks and any separately supplied explicit feedback. Ask whether the selection policy has made some categories almost invisible. A wider predeclared exposure plan is a proposal to

gather different evidence, not proof that a hidden preference was discovered. Use invented profiles and keep no-exposure states distinct from confirmed dislikes.

Log **recommender exposure**, choice and explicit feedback separately; consider whether its own policy narrows future **evidence**.

Keep the checkpoint and human choices

THIS WEEK'S CHECKPOINT Reserve some varied suggestions and distinguish 'not shown' from 'not liked.'

SYSTEM MAP Reader need → allowed **input**s → transparent rule or score → ranked candidates → **explanation** → test record → **human decision**

What will you add, change, or verify in the project this week?

> **Write or draw your reasoning and evidence.**

What should remain under **human control**?

> **Write or draw your reasoning and evidence.**

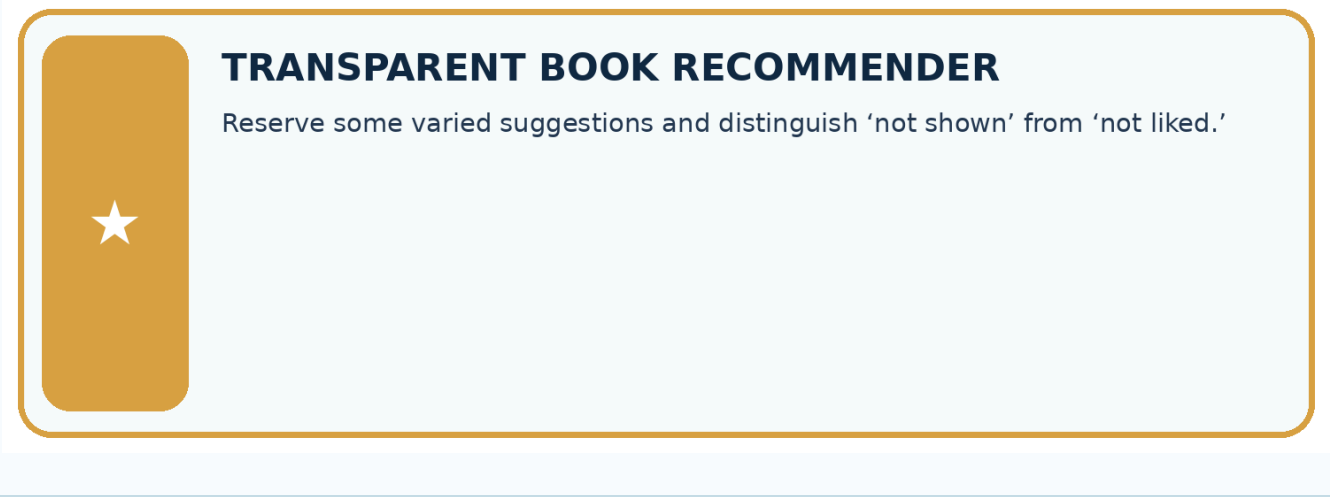
RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Transparent Book **Recommender**

Every week adds one inspectable piece. The system must show how it reached a result.



Evidence for your project

MAIN LESSON · SECTION 16

Finish: explain how selection can shape the next evidence

> Name the display count before interpreting clicks.

Your explanation and return plan

- Close your notes. Explain how selection can change exposure and the feedback available for the next recommendation.
- If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.
- Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

My deciding step, evidence and one remaining question

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	My own example	Not this / boundary
feedback loop	A cycle in which outputs affect later inputs and therefore future outputs.	
exposure	The opportunity to see or encounter an item, which can affect later choices.	
interaction data	Records of what people were shown and how they acted, such as views, clicks or corrections.	
narrowing	Reducing the allowed choices, scope or interpretation of a task.	

Terms for the connected explanation

Word	Meaning
Recommender	A procedure that suggests options using a stated policy
Tie	Two options have the same comparison score

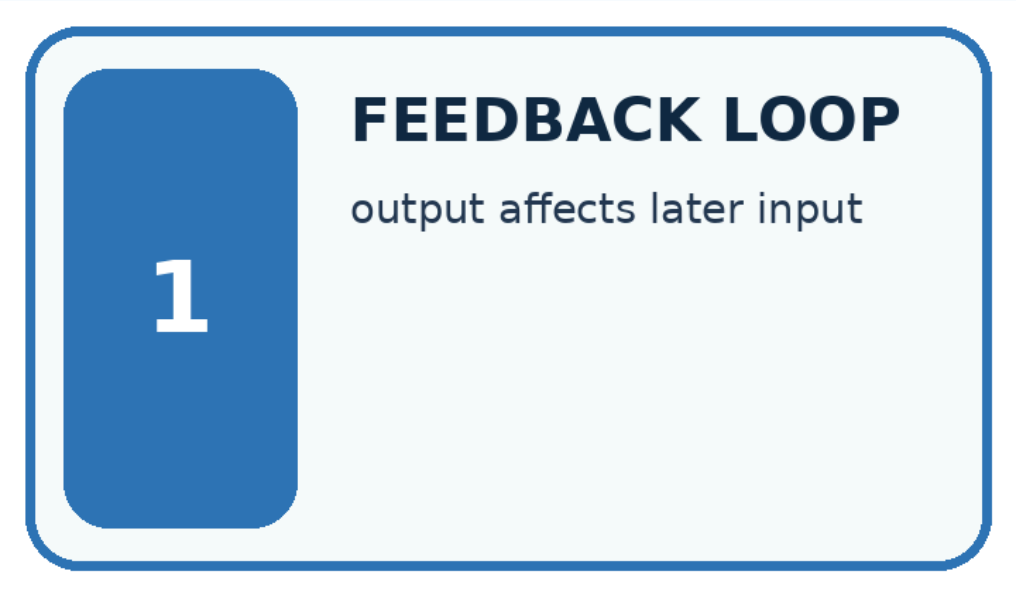
Word	Meaning
Deterministic	The same inputs and rules produce the same result
Explanation	Evidence describing how a result was obtained

▼ Earlier wording, alternatives and picture transcripts

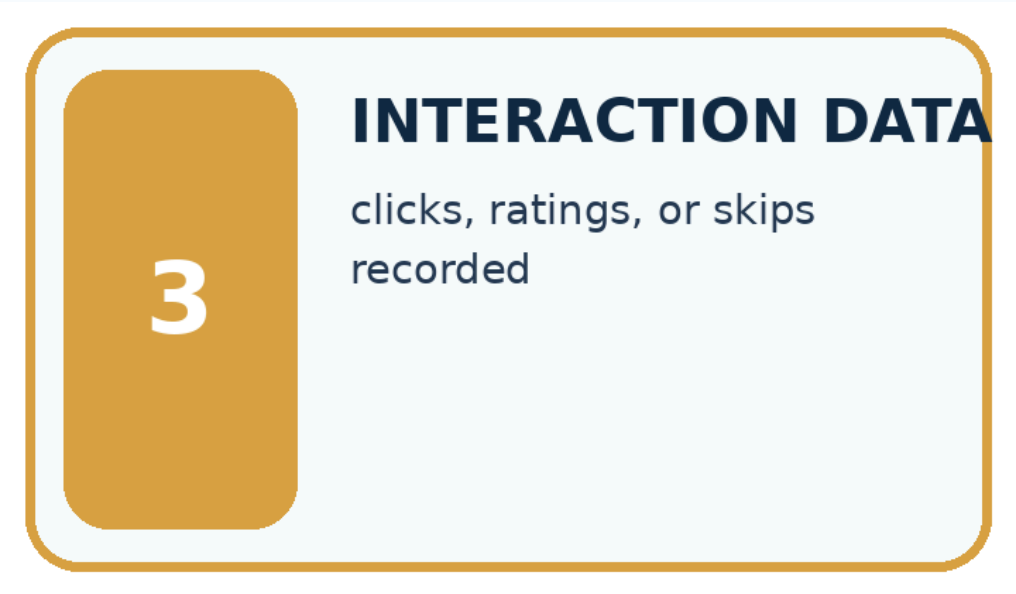
VOCABULARY LAB

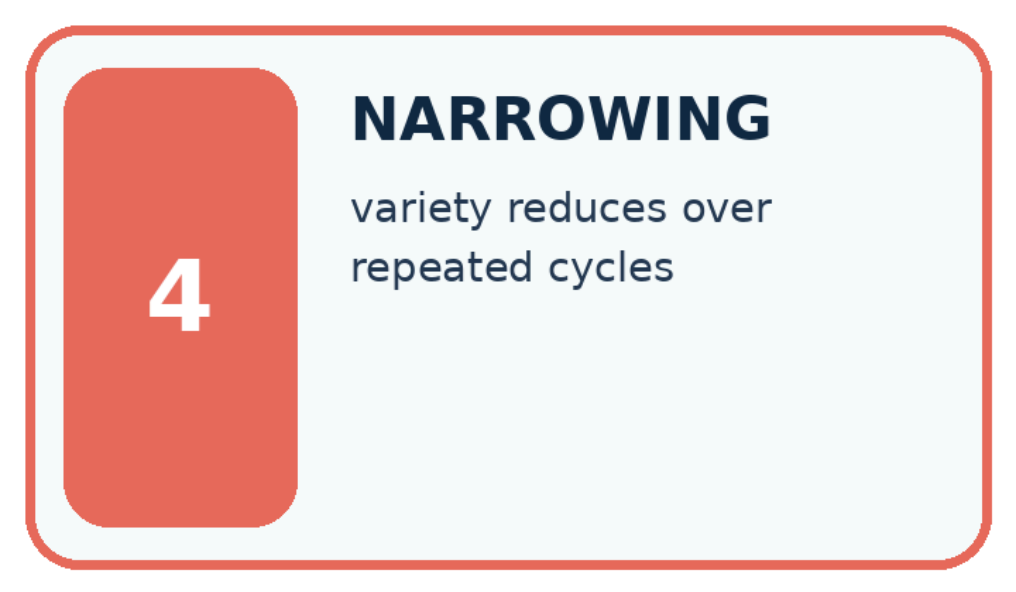
Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.




2
EXPOSURE
what the user gets a chance to see


3
INTERACTION DATA
clicks, ratings, or skips recorded



Words used in the closer explanation

▼ **Choose a stronger thinking challenge**

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

- My strongest **evidence** of independent understanding:

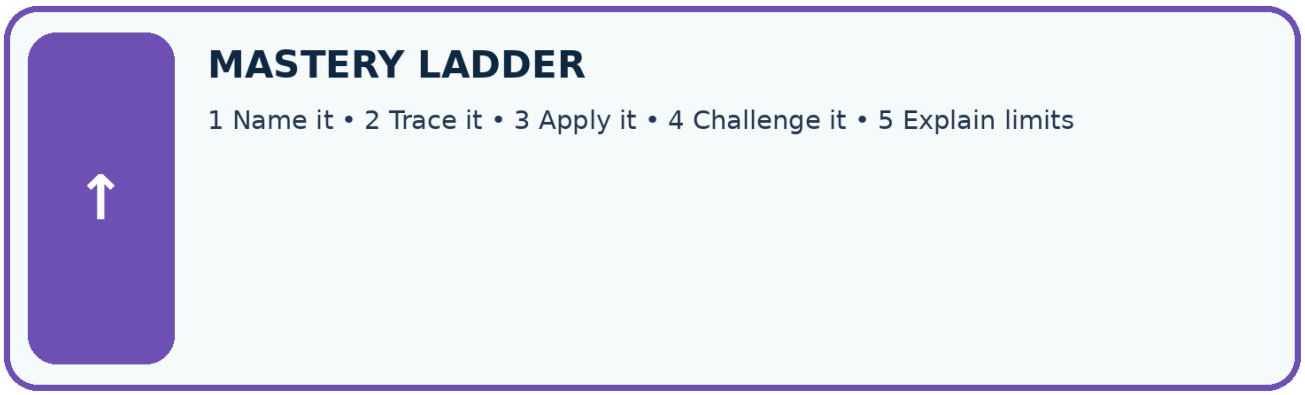
Write or draw your reasoning and evidence.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



▼ Reference: check your explanation

Reference: check your explanation

> Use after practice. Audit a new exposure log independently is the selected independent case.

Check the evidence and its limit

- 1. My fresh example, trace, calculation, or design:
- 2. My **evidence** that it works (include a boundary or missing-data case):
- 3. One limitation, fairness concern, or situation needing human review:

> **Write or draw your reasoning and evidence.**

> **Write or draw your reasoning and evidence.**

Self-check and record

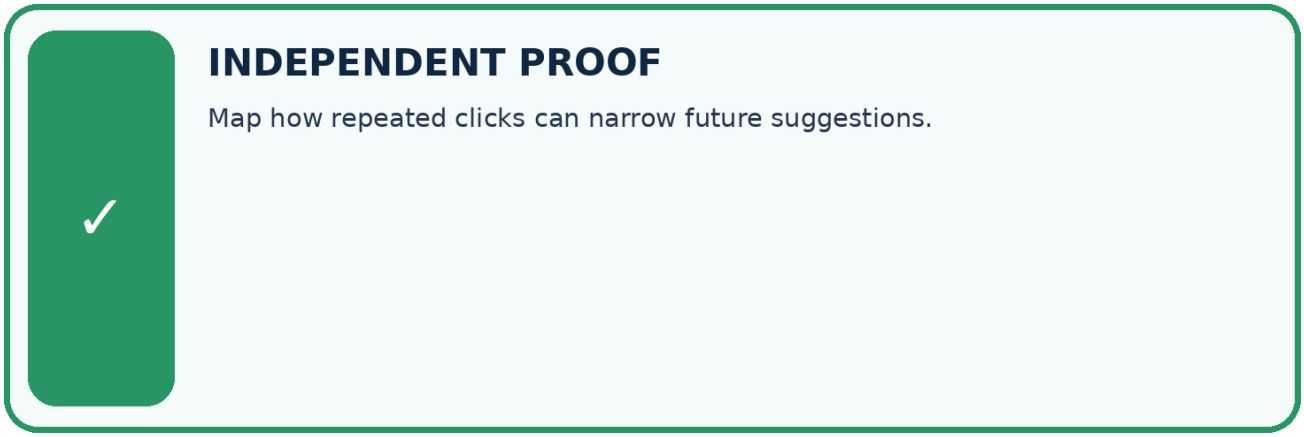
SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

▼ Earlier wording, alternatives and picture transcripts

INDEPENDENT MASTERY PROOF

Checklist for explaining this week's idea

Use after practice. Audit a new exposure log independently is the selected independent case.



EXPLANATION CHECK Keep display opportunities separate from clicks, describe the feedback mechanism and limit preference claims.

Write or draw your reasoning and evidence.

▼ **Extra practice: Exposure affects the feedback you collect**

REFERENCE · SECTION 13

Extra practice: Exposure affects the feedback you collect

> Try the question before opening its worked explanation.

Read the supplied case and question

A **recommender** shows only adventure books, uses clicks to learn and then shows more adventure.

Explain the **feedback loop**.

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

Exposure affects clicks; clicks affect future **exposure**. Record what was shown and offer controlled variety.

Evidence or stage	Value or result
Show adventure	Limits visible choices
Observe clicks	Only among shown items
Update preferences	Uses those clicks
Show more adventure	Repeats the exposure pattern

▼ Earlier wording, alternatives and picture transcripts

Week 24 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Exposure affects the feedback you collect

A **recommender** shows only adventure books, uses clicks to learn and then shows more adventure.

Explain the **feedback loop**.

Exposure affects clicks; clicks affect future **exposure**. Record what was shown and offer controlled variety.

▼ Reference: original lesson checklist

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Implement a deterministic recommender	Explain the deciding step, show a trace or test, and state one limit.
Handle ties and no eligible options	Explain the deciding step, show a trace or test, and state one limit.
Explain recommendations with their limits	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: A fictional catalogue repeatedly recommends two books using observed clicks.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional catalogue repeatedly recommends two books using observed clicks.

- Which labelled step matches this detail? B's zero may mean no **exposure**, not dislike.
- 1. Record event
- 2. Repeat selection
- 3. Initial tie
- 4. Interpret carefully
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> ---
>
> ---
>
> ---
>
> ---

▼ **Check the first explanation**

After your attempt, compare with the supported **explanation**: Interpret carefully B's zero may mean no **exposure**, not dislike. The selection policy changes which **evidence** is available.

A accumulates clicks partly because the system gives it every opportunity, while B has none. These records cannot compare enjoyment of two equally exposed books.

Keep display counts separate from click counts; investigate alternatives with a teacher-approved **exposure** plan rather than treating zero as negative feedback.

Changed-case question

- Changed-case practice: On a planned trial, show B once and record no click. Which conclusion is supported by the supplied facts?

- 1. B gains one **exposure** but still zero clicks.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

▼ **Check the changed case**

After your attempt, compare with the supported **explanation**: B gains one **exposure** but still zero clicks. The **observation** now supports a nonclick on that display, not a permanent dislike.

Repair and save your work

Review Recommendation Scores (ai-2/23). Goal: Compute a weighted score. Short practice: Why does B not become the winner when its score is largest?. Return to this week and explain what you changed.

Review Similarity by Selected Traits (ai-2/21). Goal: Represent items with numeric features. Short practice: Explain why B ceases to be a closest option without any candidate data changing..

Return to this week and explain what you changed.

Review Missing Data (ai-2/18). Goal: Detect absent and unusable values. Short practice: Why is the Boolean True rejected even though arithmetic can treat it like 1 in Python?. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work
</details>

<details>
<summary>Reference: the six-picture journey</summary>
</details>

REFERENCE · SECTION 18

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Feedback Loops: the case



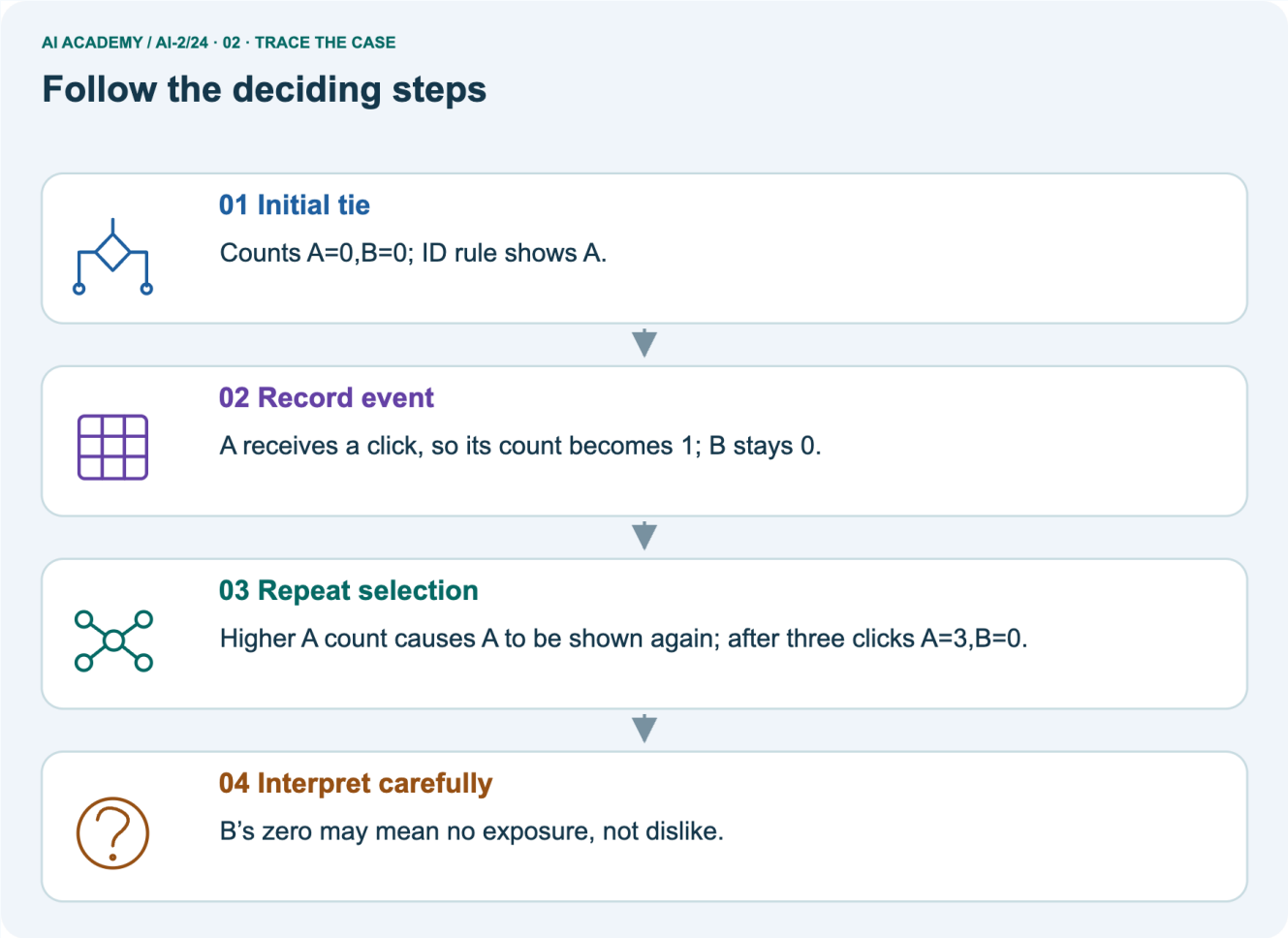
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional catalogue repeatedly recommends two books using observed clicks. Supplied facts: Initially each click count is 0. Equal counts choose earlier ID A over B. Only the recommended book is shown.

In an invented three-round record, A is shown and clicked each round; B is never shown.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

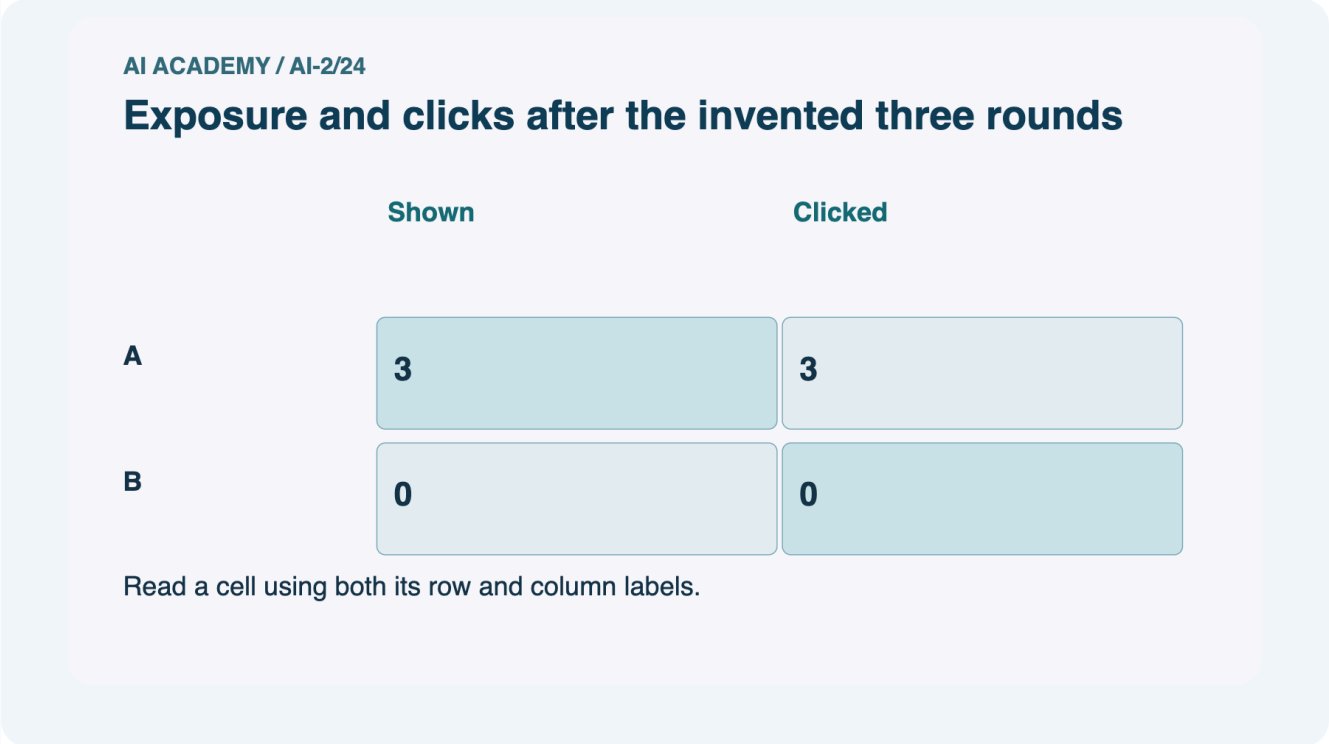


Follow the labelled picture with the matching explanation. Open the vector diagram

Initial tie: Counts A=0,B=0; ID rule shows A. → Record event: A receives a click, so its count becomes 1; B stays 0. → Repeat selection: Higher A count causes A to be shown again; after three clicks A=3,B=0. → Interpret carefully: B's zero may mean no **exposure**, not dislike.

Point to each numbered stage and explain which recorded **input** it uses.

03. Exposure and clicks after the invented three rounds

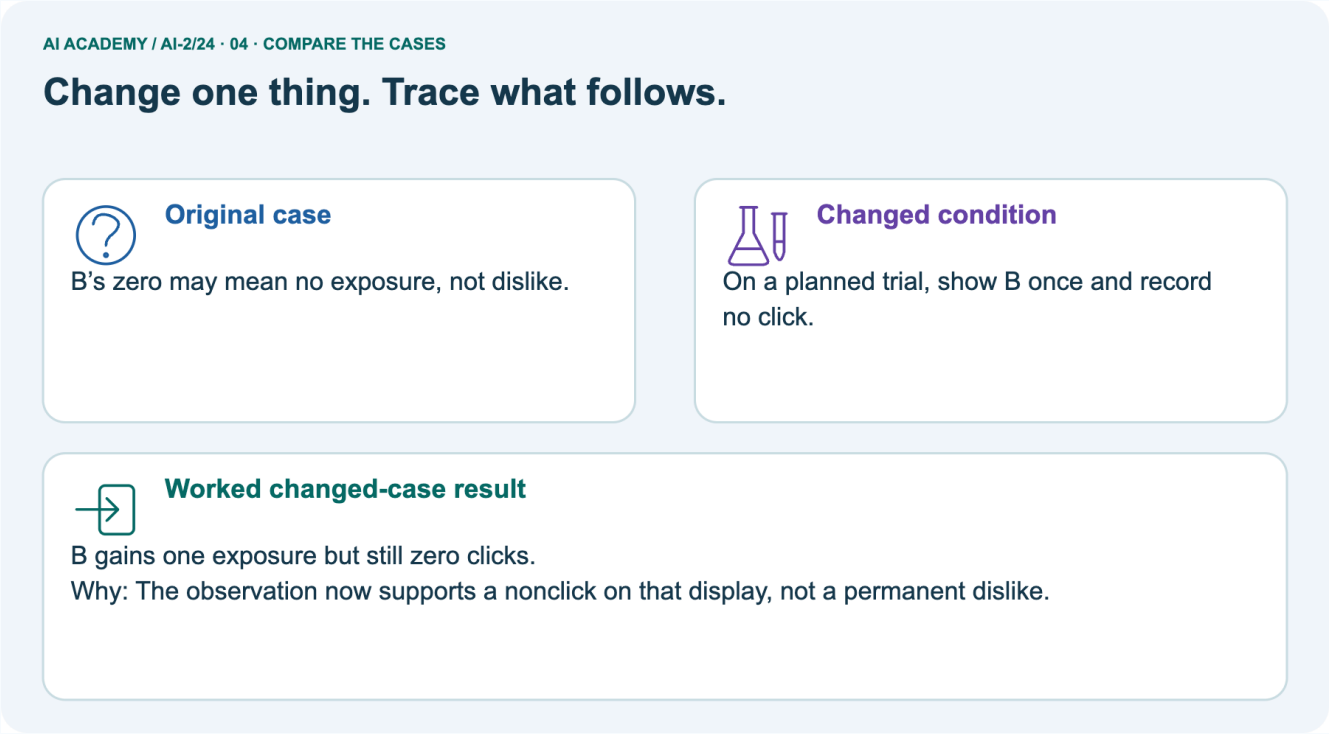


Follow the labelled picture with the matching explanation. Open the vector diagram

Exposure and clicks after the invented three rounds. A: Shown = 3, Clicked = 3. B: Shown = 0, Clicked = 0.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



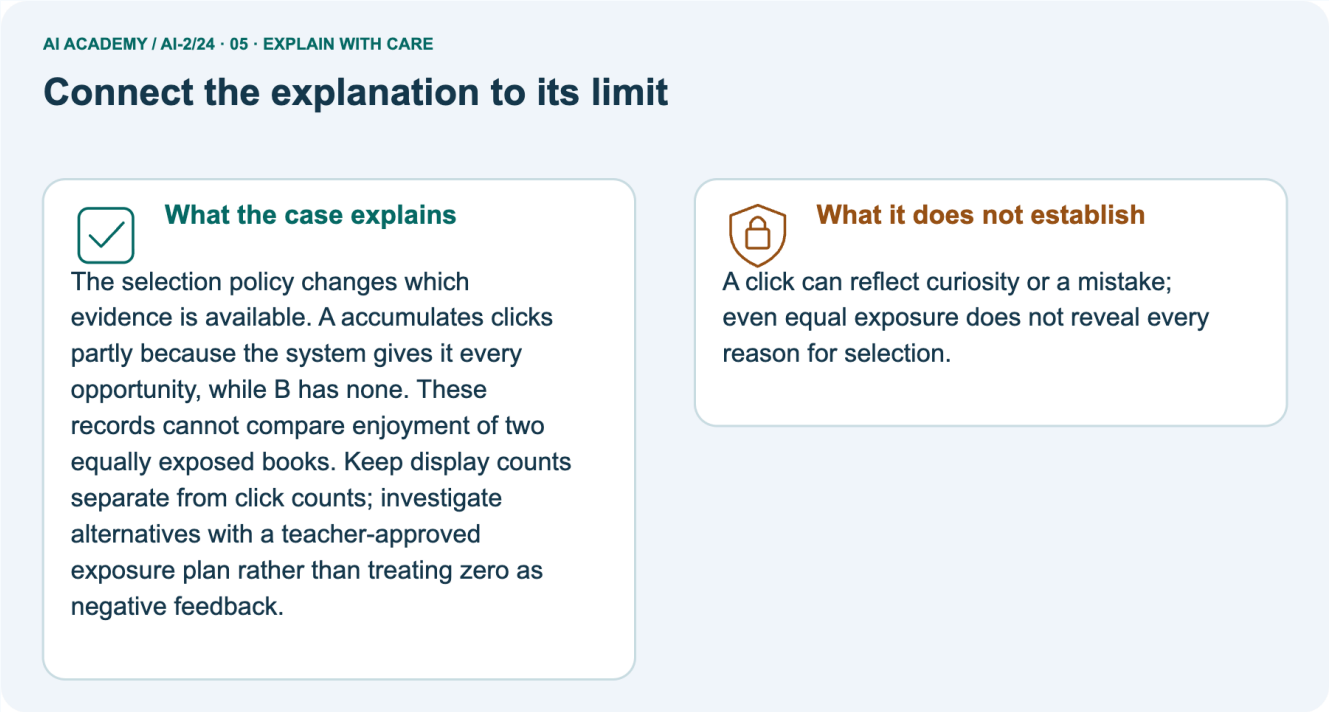
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: B's zero may mean no **exposure**, not dislike. Changed condition: On a planned trial, show B once and record no click. Expected result: B gains one **exposure** but still zero clicks.

Reason: The **observation** now supports a nonclick on that display, not a permanent dislike.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

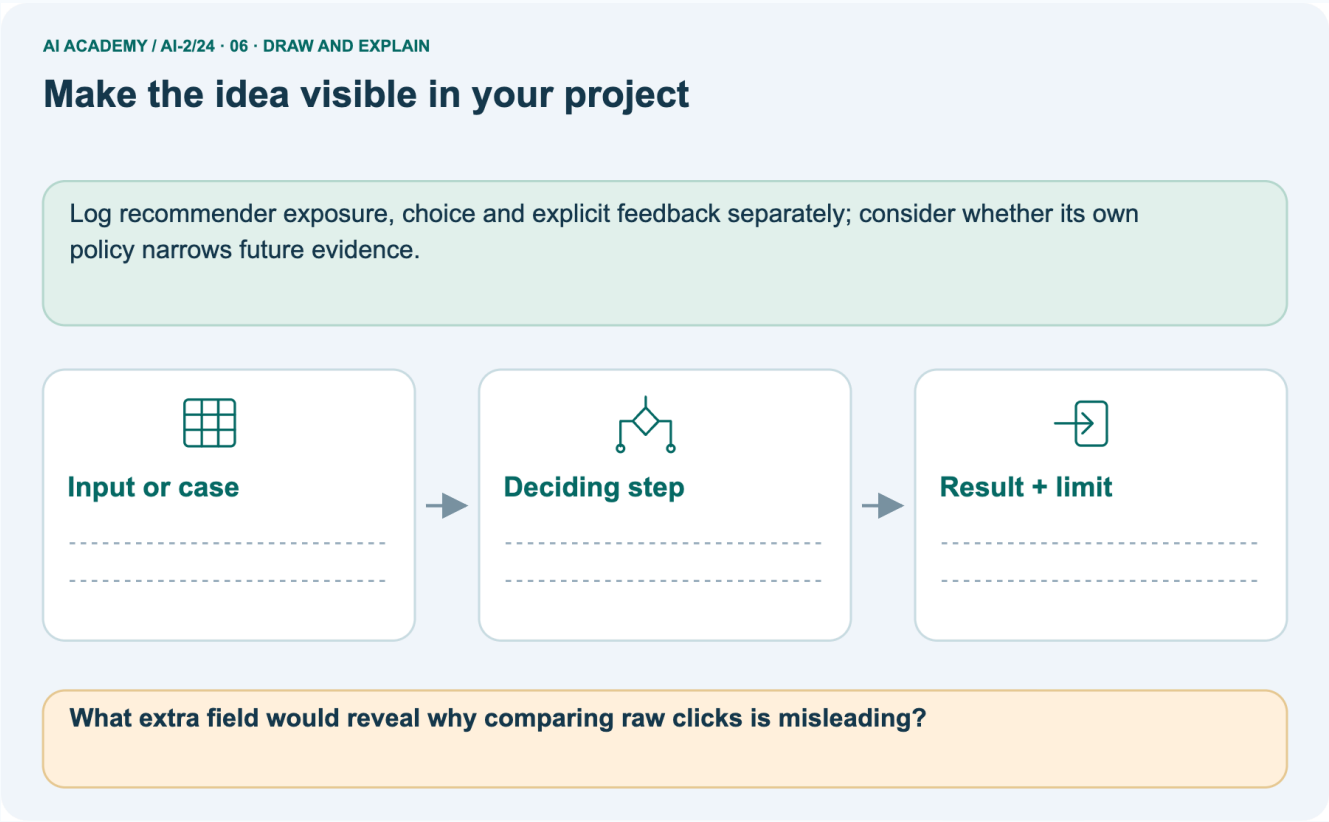
The selection policy changes which **evidence** is available. A accumulates clicks partly because the system gives it every opportunity, while B has none. These records cannot compare enjoyment of two equally exposed books.

Keep display counts separate from click counts; investigate alternatives with a teacher-approved **exposure** plan rather than treating zero as negative feedback.

Limit: A click can reflect curiosity or a mistake; even equal **exposure** does not reveal every reason for selection.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Log **recommender exposure**, choice and explicit feedback separately; consider whether its own policy narrows future **evidence**. Draw your **input**, deciding step, result and limit.

Supported question: What extra field would reveal why comparing raw clicks is misleading?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Feedback Loops: the case

02. Follow the deciding steps

03. Exposure and clicks after the invented three rounds

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.